WIGNER PHASE-SPACE REPRESENTATION, SHIFTED CAUCHY CONTOURS, AND LOCAL SADDLE-POINT COALESCENCE IN THE TRANSVERSE WRONSKIAN OF RIEMANN'S Ξ-FUNCTION

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Abstract. We study the phase-space and complex-contour structure of the transverse Wronskian $W(d,t) = \frac{1}{2}\partial_d|\Xi(t-\mathrm{i}d)|^2$ of Riemann's completed zeta function $\Xi(t-\mathrm{i}d) = \xi(\frac{1}{2}+d+\mathrm{i}t)$. First, we express $W(d,t) = 16\int_0^\infty y\sinh(2dy)\,I(y,t)\,\mathrm{d}y$ in terms of the canonical Wigner distribution $I(y,t) = 2\pi\mathcal{W}_\Phi(y/2,2t)$ of the Jacobi–Riemann kernel $\Phi(u)$ and show via Hudson's theorem and the zero-marginal identity $\int_0^\infty I(y,\gamma_k)\,\mathrm{d}y = 0$ that $I(y,t)$ changes sign. Second, we prove holomorphic parity $\Phi(-u) = \Phi(u)$ on $|\operatorname{Im}(u)| < \pi/8$ and double-exponential vertical-wall decay on $|\operatorname{Im}(z)| < \pi/4$, showing that horizontal contour shifts preserve $I(y,t)$ by Cauchy's theorem while improving L^1 quadrature conditioning. Third, we derive the local pitchfork normal form for stationary points of $\ln G_y(z) + 2\mathrm{i}tz$ away from zeros of $\Phi(u)$, locate explicit zeros of $\Phi(u)$ inside $|\operatorname{Im}(u)| < \pi/8$, and document why this formalism applies equally to functions with off-critical zeros.

1. Introduction

Let $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ denote Riemann's completed zeta function, and set $s = \frac{1}{2}+d+\mathrm{i}t$ with transverse offset $d = \sigma - \frac{1}{2} \in \mathbb{R}$ and spectral height $t \in \mathbb{R}$. Writing $z = -\mathrm{i}(s-\frac{1}{2}) = t - \mathrm{i}d$, Riemann's entire function $\Xi(z) := \xi(\frac{1}{2}+\mathrm{i}z) = R(d,t) + \mathrm{i}I_{\mathrm{im}}(d,t)$ admits the classical Fourier–Laplace representation [12, 11, 14]

$$\Xi(t-\mathrm{i}d) = 8\int_0^\infty \Phi(u)\cos\bigl(2(t-\mathrm{i}d)u\bigr)\,\mathrm{d}u = 4\int_{-\infty}^\infty \Phi(u)\,\mathrm{e}^{2(d+\mathrm{i}t)u}\,\mathrm{d}u, \tag{1.1}$$

where the rapidly decaying kernel $\Phi(u)$ is generated by the Jacobi theta series $\theta(w) = \sum_{n\in\mathbb{Z}}\mathrm{e}^{-\pi n^2 w}$ $(\operatorname{Re} w > 0)$ via

$$\Phi(u) := \sum_{n=1}^\infty \phi_n(u), \qquad \phi_n(u) := \Bigl(2\pi^2 n^4\mathrm{e}^{9u} - 3\pi n^2\mathrm{e}^{5u}\Bigr)\exp\Bigl(-\pi n^2\mathrm{e}^{4u}\Bigr). \tag{1.2}$$

By the function-theoretic positivity criteria of Hinkkanen [5] and Lagarias [8] (building on Speiser's geometric criterion for $\zeta'(s)$ [13]), the Riemann Hypothesis is equivalent to the non-negativity of $\operatorname{Re}(\xi'(s)/\xi(s)) = W(d,t)/|\Xi(t-\mathrm{i}d)|^2$ throughout $d > 0$, where the *transverse Wronskian* is defined by

$$W(d,t) := \frac{1}{2}\frac{\partial}{\partial d}|\Xi(t-\mathrm{i}d)|^2 = R\,\partial_d R + I_{\mathrm{im}}\,\partial_d I_{\mathrm{im}} = R\,\partial_t I_{\mathrm{im}} - I_{\mathrm{im}}\,\partial_t R. \tag{1.3}$$

When (1.3) is expanded directly as an oscillatory integral along the real axis, numerical quadrature faces severe cancellation: at $t = \gamma_1 \approx 14.13472514$ and shift $y = 0.40$, positive and negative cosine lobes cancel to 99.7697%. At the same time, it is essential not to conflate quadrature conditioning with a proof of positivity: by Cauchy's integral theorem, shifting a horizontal contour leaves the holomorphic integral mathematically unchanged, and the exact autocorrelation identity (1.5) holds for any real-even kernel Φ, including kernels whose Fourier transforms possess off-axis zeros (such as Davenport–Heilbronn linear combinations [3, 14] or backward de Bruijn–Newman deformations H_λ with $\lambda < 0$ [4, 9, 2]).

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This note separates three layers with explicit hypotheses: (i) unconditional exact identities for the Wigner autocorrelation representation and shifted Cauchy contours in $|\operatorname{Im}(z)| < \pi/4$; (ii) a conditional local normal form for complex stationary-point coalescence on open subsets free of zeros of Φ, illustrated numerically at $\omega = 2\gamma_1$; and (iii) explicit analytic obstructions inside the strip (zeros of $\Phi(u)$ creating logarithmic branch points in $\ln G_y(z)$, multi-crossing sign oscillations of $I(y, \gamma_k)$ for $k \geq 3$, and non-multiplicative counterexamples).

Theorem 1.1 (Canonical Wigner Identity, Shifted Cauchy Contours, and Local Saddle Normal Form)**.** *Let $d \in \mathbb{R}$ and $t \in \mathbb{R}$, and define mirror coordinates $x = u + v$, $y = u - v$ with mirror autocorrelation kernel $G_y(x) := \Phi\left(\frac{x+y}{2}\right)\Phi\left(\frac{x-y}{2}\right)$. Then:*

(i) ***Canonical Wigner representation and zero-marginal law:*** *Let $\mathcal{W}_\psi(q,p) := (2\pi)^{-1}\int_{\mathbb{R}} \psi(q+x/2)\overline{\psi(q-x/2)}\,\mathrm{e}^{-\mathrm{i}px}\,\mathrm{d}x$ denote the standard quantum-mechanical Wigner distribution [15, 6]. Defining the inner correlation integral*

$$I(y,t) := \int_{-\infty}^{\infty} G_y(x)\cos(2tx)\,\mathrm{d}x = 2\pi\,\mathcal{W}_\Phi\left(\frac{y}{2},\,2t\right), \tag{1.4}$$

one has $|\Xi(t-\mathrm{id})|^2 = 16\int_0^\infty \cosh(2dy)\,I(y,t)\,\mathrm{d}y$, $\int_0^\infty I(y,t)\,\mathrm{d}y = \frac{1}{16}|\Xi(t)|^2$, and

$$W(d,t) = 16\int_0^\infty y\sinh(2dy)\,I(y,t)\,\mathrm{d}y, \qquad \widetilde{W}(d,t) := \frac{W(d,t)}{d} = 32\int_0^\infty y^2\,\frac{\sinh(2dy)}{2dy}\,I(y,t)\,\mathrm{d}y. \tag{1.5}$$

At every real zero γ_k of Ξ, $\int_0^\infty I(y,\gamma_k)\,\mathrm{d}y = 0$, and since Φ is non-Gaussian, Hudson's theorem [6] implies that $I(y,t)$ changes sign in $(y,t) \in \mathbb{R}^2$.

(ii) ***Holomorphic strip, Cauchy contour invariance, and strip zeros of Φ:*** *The defining series (1.2) converges locally uniformly on $|\operatorname{Im}(u)| < \pi/8$, where $\Phi(-u) = \Phi(u)$ holds holomorphically, and ceases to converge on $|\operatorname{Im}(u)| = \pi/8$. Hence for each $y \in \mathbb{R}$, $z \mapsto G_y(z)$ is holomorphic and even on $\mathcal{S}_{\pi/4} := \{z = \xi + \mathrm{i}\eta \in \mathbb{C} : |\eta| < \pi/4\}$, decays double-exponentially as $|G_y(\pm L + \mathrm{i}\eta)| \leq 16\pi^4\mathrm{e}^{9L}\exp(-2\pi\cosh(2y)\mathrm{e}^{2L}\cos(2\eta))$ as $L \to \infty$, and satisfies $G_y(-\xi+\mathrm{i}\eta) = \overline{G_y(\xi+\mathrm{i}\eta)}$ with $G_y(\mathrm{i}\eta) = |\Phi((y+\mathrm{i}\eta)/2)|^2 \geq 0$. For every $\eta \in (0,\pi/4)$ and $t > 0$, Cauchy's theorem gives the contour-invariant identity*

$$I(y,t) = \mathrm{e}^{-2t\eta}\int_{-\infty}^{\infty} \operatorname{Re}\left[G_y(\xi+\mathrm{i}\eta)\,\mathrm{e}^{2\mathrm{i}t\xi}\right]\mathrm{d}\xi. \tag{1.6}$$

However, $\Phi(u)$ is not *zero-free on $|\operatorname{Im}(u)| < \pi/8$: in particular, $\Phi(\mathrm{i}\tau_0) = 0$ at $\tau_0 \approx 0.3194150268 < \pi/8$ and $\Phi(u_1) = 0$ at $u_1 \approx 0.1278237567 + 0.3477699161\mathrm{i}$, producing logarithmic branch points of $\ln G_y(z)$ inside $\mathcal{S}_{\pi/4}$.*

(iii) ***Conditional local pitchfork coalescence normal form:*** *Let $\omega = 2t > 0$, and suppose $(y_c,\eta_c) \in (0,\infty)\times(0,\pi/4)$ satisfies $G_{y_c}(\mathrm{i}\eta_c) > 0$ together with*

$$L'_{y_c}(\eta_c) = \omega, \qquad L''_{y_c}(\eta_c) = 0, \qquad \partial_y L'_{y_c}(\eta_c) \neq 0, \qquad L'''_{y_c}(\eta_c) \neq 0,$$

where $L_y(\eta) := \ln G_y(\mathrm{i}\eta)$. Under the sign conditions $\partial_y L'_{y_c}(\eta_c) > 0$ and $L'''_{y_c}(\eta_c) < 0$, the stationary-point equation $\partial_z \ln G_y(z) + \mathrm{i}\omega = 0$ undergoes a local supercritical pitchfork bifurcation in a neighborhood of $(y_c, \mathrm{i}\eta_c)$: two symmetric off-axis roots $z_*^\pm(y,\omega) = \pm\xi_*(y,\omega) + \mathrm{i}\eta_*(y,\omega)$ for $y < y_c$ (y near y_c) merge onto a single local imaginary-axis root $z_*(y,\omega) = \mathrm{i}\eta_*(y,\omega)$ with $\partial_\xi^2 \ln G_y(\mathrm{i}\eta_*) < 0$ for $y > y_c$ (y near y_c).

The numerical constants verified at 50–60 decimal places in Section 5 at the first Riemann zero $\gamma_1 \approx 14.1347251417$ ($\omega_1 = 2\gamma_1 \approx 28.2694502835$) are:

$$\begin{aligned} \eta_{\max} &= \frac{\pi}{4} \approx 0.7853981634, & \tau_0 &\approx 0.3194150268 \;\; (2\tau_0 \approx 0.6388300536), \\ y_c(\omega_1) &\approx 0.4320205303, & \eta_c(\omega_1) &\approx 0.6027150239, \\ \kappa_{\mathrm{quad}}(\eta = 0) &\approx 867.6007 \;\; (y = 0.40), & \kappa_{\mathrm{quad}}(\eta = 0.50) &\approx 1.0008 \;\; (y = 0.40). \end{aligned} \tag{1.7}$$

Table 1 separates classical results in the literature from the identities, local analysis, and obstructions recorded in this note.

TABLE 1. Relationship between classical results and the analysis in this note.

Author(s)	**Established Classical Result**	**Content of Present Note**
Riemann (1859) [12], Pólya (1926) [11]	Representation of $\Xi(z)$ via the Jacobi-theta kernel $\Phi(u)$.	Mirror-coordinate formula $W(d,t) = 16\int_0^\infty y\sinh(2dy)I(y,t)\,\mathrm{d}y$ with $I(y,t) = 2\pi\mathcal{W}_\Phi(y/2,2t)$.
Speiser (1934) [13], Hinkkanen (1997) [5], Lagarias (1999) [8]	Equivalence of RH with $\zeta'(s) \neq 0$ in $0 < \sigma < 1/2$ and $\mathrm{Re}(\xi'(s)/\xi(s)) > 0$ in $\sigma > 1/2$.	Parity-reduced hyperbolic-sinc representation $\widetilde{W}(d,t) = W(d,t)/d$ and documentation of DH counterexamples.
Wigner (1932) [15], Hudson (1974) [6]	$\mathcal{W}_\psi(q,p) \geq 0$ everywhere on $\mathbb{R}^2$ if and only if ψ is Gaussian.	Zero-marginal identity $\int_0^\infty I(y,\gamma_k)\,\mathrm{d}y = 0$ and multi-crossing behavior for $k \geq 3$.
Chester–Friedman–Ursell (1957) [1], Olver (1974) [10]	Local cubic normal form for two coalescing complex saddle points.	Local pitchfork normal form for $G_y(z)\mathrm{e}^{\mathrm{i}\omega z}$ away from Φ-zeros, plus explicit location of strip Φ-zeros.

1.1. **Proof overview.** The argument proceeds in four steps:

(1) *Autocorrelation and Wigner scaling (Section 2):* Applying $x = u+v$, $y = u-v$ (Jacobian $\frac{1}{2}$) to $|\Xi(t-\mathrm{i}d)|^2 = 16\iint_{\mathbb{R}^2}\Phi(u)\Phi(v)\mathrm{e}^{2d(u-v)}\cos(2t(u+v))\,\mathrm{d}u\,\mathrm{d}v$ yields $|\Xi(t-\mathrm{i}d)|^2 = 16\int_0^\infty\cosh(2dy)\,I(y,t)\,\mathrm{d}y$ and $W(d,t) = 16\int_0^\infty y\sinh(2dy)\,I(y,t)\,\mathrm{d}y$, where $G_y(x) = \Phi(\frac{y}{2}+\frac{x}{2})\Phi(\frac{y}{2}-\frac{x}{2})$ identifies $I(y,t) = 2\pi\mathcal{W}_\Phi(y/2,2t)$.

(2) *Holomorphic strip and Cauchy rectangle (Section 3):* The differential representation $\Phi(u) = \frac{1}{16}(f''(u)-f(u))$ for $f(u) = \mathrm{e}^u\theta(\mathrm{e}^{4u})$, combined with Jacobi's modular relation $\theta(\mathrm{e}^{-4u}) = \mathrm{e}^{2u}\theta(\mathrm{e}^{4u})$, proves $\Phi(-u) = \Phi(u)$ holomorphically on $|\mathrm{Im}(u)| < \pi/8$. Closing Cauchy's rectangle on $[-L,L]\times[0,\eta]$ ($\eta \in (0,\pi/4)$) yields (1.6). We then prove $\Phi(\mathrm{i}\tau)$ changes sign on $(0,\pi/8)$ and locate off-axis zeros of $\Phi(u)$ in the strip.

(3) *Local saddle coalescence (Section 4):* On open neighborhoods where $G_y(z) \neq 0$, Hermitian symmetry $G_y(-\xi+\mathrm{i}\eta) = \overline{G_y(\xi+\mathrm{i}\eta)}$ forces $\partial_\xi^k \ln G_y(\mathrm{i}\eta)$ to be purely imaginary for odd k. Expanding $\partial_z \ln G_y(z) + \mathrm{i}\omega = 0$ around $(y_c,\mathrm{i}\eta_c)$ yields the local pitchfork law $(\Delta z)^2 = K^2(y_c-y) + \mathcal{O}(|y-y_c|^{3/2})$.

(4) *Quadrature conditioning and obstructions (Sections 5–6):* High-precision computations in `mpmath` verify (1.5) to 54 decimals, quantify shifted-contour quadrature conditioning, and exhibit the multi-crossing and Davenport–Heilbronn obstructions.

1.2. **Notation and conventions.** Throughout this series, $s = \sigma+\mathrm{i}t = \frac{1}{2}+d+\mathrm{i}t$ with $d = \sigma-\frac{1}{2} \in \mathbb{R}$, $t \in \mathbb{R}$, and $z = -\mathrm{i}(s-\frac{1}{2}) = t-\mathrm{i}d$, so $\Xi(t-\mathrm{i}d) = \xi(\frac{1}{2}+d+\mathrm{i}t)$. When discussing contour integrals at a general Fourier frequency we write $\omega = 2t > 0$ and $I_{\mathrm{freq}}(y,\omega) := \int_{-\infty}^\infty G_y(x)\cos(\omega x)\,\mathrm{d}x$, so that the canonical correlation kernel in (1.4) satisfies $I(y,t) \equiv I_{\mathrm{freq}}(y,2t)$ everywhere. Complex contour coordinates in the x-plane are written $z = \xi+\mathrm{i}\eta$ with $\xi \in \mathbb{R}$ and $|\eta| < \pi/4$. For any real-valued integrand $f(\xi)$ on $\mathbb{R}$ with non-zero integral, $\kappa_{\mathrm{quad}}[f] := (\int_{\mathbb{R}}|f(\xi)|\,\mathrm{d}\xi)/|\int_{\mathbb{R}}f(\xi)\,\mathrm{d}\xi| \geq 1$ denotes its L^1 quadrature condition number.

2. Canonical Wigner representation and zero-marginal law

We begin by deriving the exact autocorrelation representation of $|\Xi(t-\mathrm{i}d)|^2$ and $W(d,t)$ with particular care for prefactors and standard Wigner scaling.

Proposition 2.1 (Mirror autocorrelation and canonical Wigner scaling)**.** *For all $(d,t) \in \mathbb{R}^2$, $|\Xi(t-\mathrm{i}d)|^2$, $W(d,t)$, and $\widetilde{W}(d,t) := W(d,t)/d$ satisfy*

$$|\Xi(t-\mathrm{i}d)|^2 = 16\int_0^\infty \cosh(2dy)\,I(y,t)\,\mathrm{d}y, \tag{2.1}$$

together with (1.4) and (1.5). In particular, at $d = 0$, $\int_0^\infty I(y,t)\,\mathrm{d}y = \frac{1}{16}|\Xi(t)|^2$.

Proof. Because $\Phi(u)$ is real and even on $\mathbb{R}$ and $\Phi(u) = \mathcal{O}(\mathrm{e}^{9|u|}\exp(-\pi\mathrm{e}^{4|u|}))$, multiplying $\Xi(t-\mathrm{i}d) = 4\int_{\mathbb{R}}\Phi(u)\mathrm{e}^{2(d+\mathrm{i}t)u}\,\mathrm{d}u$ by $\overline{\Xi(t-\mathrm{i}d)} = 4\int_{\mathbb{R}}\Phi(v)\mathrm{e}^{2(d-\mathrm{i}t)v}\,\mathrm{d}v$ and substituting $v \mapsto -v$ by Fubini's theorem gives

$$\begin{aligned}\left|\Xi(t-\mathrm{i}d)\right|^2 &= 16\iint_{\mathbb{R}^2}\Phi(u)\Phi(v)\,\mathrm{e}^{2d(u-v)}\,\mathrm{e}^{2\mathrm{i}t(u+v)}\,\mathrm{d}u\,\mathrm{d}v\\ &= 16\iint_{\mathbb{R}^2}\Phi(u)\Phi(v)\cosh\left(2d(u-v)\right)\cos\left(2t(u+v)\right)\mathrm{d}u\,\mathrm{d}v,\end{aligned}$$

where the last step symmetrizes over $(u,v)\leftrightarrow(v,u)$. Changing variables to $x = u+v$ and $y = u-v$ gives $u = \frac{x+y}{2}$, $v = \frac{x-y}{2}$, and $\mathrm{d}u\,\mathrm{d}v = \frac{1}{2}\mathrm{d}x\,\mathrm{d}y$, whence

$$\begin{aligned}\left|\Xi(t-\mathrm{i}d)\right|^2 &= 8\int_{-\infty}^{\infty}\cosh(2dy)\left(\int_{-\infty}^{\infty}\Phi\left(\frac{x+y}{2}\right)\Phi\left(\frac{x-y}{2}\right)\cos(2tx)\,\mathrm{d}x\right)\mathrm{d}y\\ &= 16\int_0^{\infty}\cosh(2dy)\,I(y,t)\,\mathrm{d}y,\end{aligned}$$

since $I(-y,t) = I(y,t)$. Because $|y\sinh(2dy)I(y,t)| \le |y|\sinh(2|d||y|)I(y,0)$ is integrable on $[0,\infty)$ uniformly on compact d-intervals, differentiating (2.1) under the integral sign gives $\partial_d|\Xi(t-\mathrm{i}d)|^2 = 32\int_0^{\infty} y\sinh(2dy)\,I(y,t)\,\mathrm{d}y$, and dividing by 2 yields $W(d,t) = 16\int_0^{\infty} y\sinh(2dy)\,I(y,t)\,\mathrm{d}y$ and $\widetilde{W}(d,t) = 16\int_0^{\infty} y\frac{\sinh(2dy)}{d}\,I(y,t)\,\mathrm{d}y = 32\int_0^{\infty} y^2\frac{\sinh(2dy)}{2dy}\,I(y,t)\,\mathrm{d}y$. Finally, since Φ is real and even, the standard Wigner distribution $\mathcal{W}_\Phi(q,p) = \frac{1}{2\pi}\int_{\mathbb{R}}\Phi(q+\frac{x}{2})\Phi(q-\frac{x}{2})\cos(px)\,\mathrm{d}x$ evaluated at $(q,p) = (y/2, 2t)$ equals $\frac{1}{2\pi}I(y,t)$, proving (1.4). □

Proposition 2.2 (Hudson sign indefiniteness and single-crossing moment identity)**.** *The correlation kernel $I(y,t)$ satisfies:*

(a) ***Zero-marginal identity and sign changes:*** *At every real zero γ_k of Ξ, the marginal integral vanishes: $\int_0^{\infty} I(y,\gamma_k)\,\mathrm{d}y = \frac{1}{16}|\Xi(\gamma_k)|^2 = 0$. Since $\Phi \in L^2(\mathbb{R})$ decays double-exponentially as $|u|\to\infty$, Φ is not Gaussian, so by Hudson's theorem [6] $\mathcal{W}_\Phi$ (and hence $I(y,t)$) assumes strictly negative values on $\mathbb{R}^2$; in particular, $I(\cdot,\gamma_k)$ cannot have constant sign on $(0,\infty)$ unless it vanishes identically, which is ruled out by Fourier uniqueness since $G_y(0) = \Phi(y/2)^2 \not\equiv 0$.*

(b) ***Subtracted moment identity:*** *For every $y_0 \ge 0$, $d > 0$, and $t \in \mathbb{R}$,*

$$W(d,t) = y_0\sinh(2dy_0)\left|\Xi(t)\right|^2 + 16\int_0^{\infty}\left(y\sinh(2dy) - y_0\sinh(2dy_0)\right)I(y,t)\,\mathrm{d}y. \tag{2.2}$$

If, at a given t, $I(y,t)$ happens to have a single *sign change from ≤ 0 on $[0,y_0)$ to ≥ 0 on (y_0,∞) (not identically zero), then (2.2) immediately implies $W(d,t) > 0$ for all $d > 0$ [7].*

Proof. Part (a) follows from $\int_0^{\infty} I(y,\gamma_k)\,\mathrm{d}y = \frac{1}{16}|\Xi(\gamma_k)|^2 = 0$ and Hudson's characterization [6] of non-negative Wigner distributions. Subtracting $16y_0\sinh(2dy_0)\int_0^{\infty} I(y,t)\,\mathrm{d}y = y_0\sinh(2dy_0)|\Xi(t)|^2$ from $W(d,t) = 16\int_0^{\infty} y\sinh(2dy)I(y,t)\,\mathrm{d}y$ gives (2.2); since $y\mapsto y\sinh(2dy)$ is strictly increasing on $[0,\infty)$ for $d > 0$, $(y\sinh(2dy) - y_0\sinh(2dy_0))I(y,t) \ge 0$ everywhere under the single-crossing hypothesis. □

Remark 2.3 (Failure of single-crossing for $k \ge 3$). As verified numerically in Table 4, the single-crossing hypothesis in Proposition 2.2(b) holds at γ_1 and γ_2, where $I(y,\gamma_k)$ crosses zero once on $(0, 1.35]$, but *fails already at* $\gamma_3 \approx 25.010858$, where $I(y,\gamma_3)$ exhibits 5 sign changes (6 at γ_4, 7 at γ_5). Thus (2.2) does not provide a general positivity proof for $W(d,\gamma_k)$.

3. Holomorphic strip, Cauchy contour shift, and strip zeros of Φ

To deform the x-integral into the complex plane $z = \xi + \mathrm{i}\eta$, one must establish holomorphy of $\Phi(u)$ without real-axis absolute values, justify the vertical Cauchy walls, and locate the zeros of $\Phi(u)$ inside the convergence strip.

Lemma 3.1 (Holomorphic parity on $|\mathrm{Im}(u)| < \pi/8$)**.** *The series (1.2) defines a holomorphic function $\Phi(u)$ on the open strip $\mathcal{U}_{\pi/8} := \{u = \sigma + \mathrm{i}\tau \in \mathbb{C} : |\tau| < \pi/8\}$ satisfying $\Phi(-u) = \Phi(u)$ and $\Phi(\bar{u}) = \overline{\Phi(u)}$ for all $u \in \mathcal{U}_{\pi/8}$.*

Proof. For $u = \sigma + \mathrm{i}\tau$ with $|\tau| < \pi/8$, $\mathrm{Re}(\mathrm{e}^{4u}) = \mathrm{e}^{4\sigma}\cos(4\tau) > 0$. On any compact $K \subset \mathcal{U}_{\pi/8}$, $\sigma \geq \sigma_0$ and $\cos(4\tau) \geq \delta_0 > 0$, so $|\exp(-\pi n^2 \mathrm{e}^{4u})| \leq \exp(-\pi n^2 \mathrm{e}^{4\sigma_0}\delta_0)$, giving uniform convergence of (1.2) by Weierstrass's theorem. Setting $f(u) := \mathrm{e}^u\theta(\mathrm{e}^{4u})$ for $\mathrm{Re}(\mathrm{e}^{4u}) > 0$, direct differentiation of $f(u) = \mathrm{e}^u + 2\sum_{n=1}^{\infty}\mathrm{e}^u\exp(-\pi n^2\mathrm{e}^{4u})$ yields $\frac{1}{16}(f''(u) - f(u)) = \Phi(u)$. Jacobi's theta transformation $\theta(1/w) = w^{1/2}\theta(w)$ (principal branch on $\mathrm{Re}\, w > 0$) evaluated at $w = \mathrm{e}^{4u}$ gives $\theta(\mathrm{e}^{-4u}) = \mathrm{e}^{2u}\theta(\mathrm{e}^{4u})$, so $f(-u) = \mathrm{e}^{-u}\theta(\mathrm{e}^{-4u}) = \mathrm{e}^u\theta(\mathrm{e}^{4u}) = f(u)$. Differentiating twice gives $\Phi(-u) = \Phi(u)$, and real coefficients on $\mathbb{R}$ give $\Phi(\bar{u}) = \overline{\Phi(u)}$. □

Proposition 3.2 (Vertical wall decay, Hermitian symmetry, and strip zeros of Φ)**.** *Fix $y \in \mathbb{R}$ and set $G_y(z) := \Phi\left(\frac{z+y}{2}\right)\Phi\left(\frac{z-y}{2}\right)$ for $z = \xi + \mathrm{i}\eta \in \mathcal{S}_{\pi/4} = \{|\eta| < \pi/4\}$. Then:*

(a) ***Double-exponential wall decay and contour invariance:*** *For every $\eta \in [0, \pi/4)$ and all sufficiently large $L > |y| + 1$,*

$$\sup_{|\tau|\leq\eta}|G_y(\pm L + \mathrm{i}\tau)| \leq 16\pi^4\mathrm{e}^{9L}\exp\Big(-2\pi\cosh(2y)\,\mathrm{e}^{2L}\cos(2\eta)\Big), \tag{3.1}$$

so the vertical wall integrals vanish as $L \to \infty$ and (1.6) holds for every $\eta \in (0, \pi/4)$ and $t > 0$. On $|\eta| = \pi/4$, $\cos(2\eta) = 0$ and the defining series (1.2) ceases to converge.

(b) ***Hermitian reflection symmetry:*** *For all $\xi \in \mathbb{R}$ and $|\eta| < \pi/4$, $G_y(-\xi + \mathrm{i}\eta) = \overline{G_y(\xi + \mathrm{i}\eta)}$, and on the imaginary axis $G_y(\mathrm{i}\eta) = \left|\Phi(\frac{y+\mathrm{i}\eta}{2})\right|^2 \geq 0$.*

(c) ***Existence of zeros of $\Phi(u)$ inside $\mathcal{U}_{\pi/8}$:*** *The kernel $\Phi(u)$ vanishes inside $\mathcal{U}_{\pi/8}$. Specifically, on the imaginary axis $u = \mathrm{i}\tau$ ($\tau \in [0, \pi/8)$), $\Phi(\mathrm{i}\tau)$ is real-valued, satisfies $\Phi(0) > 0$, and tends to 0^- as $\tau \to (\pi/8)^-$, vanishing at $\tau_0 = 0.31941502680029678874\cdots < \pi/8$; hence $G_0(\mathrm{i}\eta_0) = 0$ at $\eta_0 = 2\tau_0 \approx 0.6388300536 < \pi/4$. Moreover, $\Phi(u)$ also has off-axis zeros in $\mathcal{U}_{\pi/8}$, such as $u_1 \approx 0.1278237567 + 0.3477699161\mathrm{i}$, producing zeros of $G_y(z)$ at $z = \pm 2u_1 \pm y$ with imaginary height $\eta_1 = 2\,\mathrm{Im}(u_1) \approx 0.6955398322 < \pi/4$.*

Proof. (a) For $z = L + \mathrm{i}\tau$ with $0 \leq |\tau| \leq \eta < \pi/4$ and $L > |y| + 1$, let $u_\pm = \frac{L\pm y}{2} + \mathrm{i}\frac{\tau}{2}$ and $\sigma_\pm = \frac{L\pm y}{2} > \frac{1}{2}$. Since $\cos(2\tau) \geq \cos(2\eta) > 0$ and $2\pi^2 n^4\mathrm{e}^{9\sigma_\pm} > 3\pi n^2\mathrm{e}^{5\sigma_\pm}$, bounding the $n \geq 2$ tail of (1.2) geometrically yields $|\Phi(u_\pm)| \leq 4\pi^2\mathrm{e}^{9\sigma_\pm}\exp(-\pi\mathrm{e}^{4\sigma_\pm}\cos(2\tau))$ for L large enough that the tail is at most the $n = 1$ bound. Multiplying $|\Phi(u_+)|$ and $|\Phi(u_-)|$ and using $\mathrm{e}^{2(L+y)} + \mathrm{e}^{2(L-y)} = 2\cosh(2y)\mathrm{e}^{2L}$ gives (3.1); $G_y(-z) = G_y(z)$ covers $-L + \mathrm{i}\tau$. Since $I(y,t) = \mathrm{Re}\int_{\mathbb{R}} G_y(\xi)\mathrm{e}^{2\mathrm{i}t\xi}\,\mathrm{d}\xi$ and $z \mapsto G_y(z)\mathrm{e}^{2\mathrm{i}tz}$ is holomorphic on $\mathcal{S}_{\pi/4}$ (zeros of G_y do not affect holomorphy of $G_y(z)\mathrm{e}^{2\mathrm{i}tz}$), Cauchy's theorem on $[-L, L] \times [0, \eta]$ as $L \to \infty$ proves (1.6).

(b) Applying $\Phi(-u) = \Phi(u)$ and $\Phi(\bar{u}) = \overline{\Phi(u)}$ gives

$$G_y(-\xi+\mathrm{i}\eta) = \Phi\left(\frac{-\xi+y+\mathrm{i}\eta}{2}\right)\Phi\left(\frac{-\xi-y+\mathrm{i}\eta}{2}\right) = \Phi\left(\overline{\frac{\xi-y+\mathrm{i}\eta}{2}}\right)\Phi\left(\overline{\frac{\xi+y+\mathrm{i}\eta}{2}}\right) = \overline{G_y(\xi+\mathrm{i}\eta)},$$

and setting $\xi = 0$ gives $G_y(\mathrm{i}\eta) = \Phi(\frac{y+\mathrm{i}\eta}{2})\overline{\Phi(\frac{y+\mathrm{i}\eta}{2})} = |\Phi(\frac{y+\mathrm{i}\eta}{2})|^2 \geq 0$.

(c) By Lemma 3.1, $\Phi(-\mathrm{i}\tau) = \Phi(\mathrm{i}\tau) = \overline{\Phi(\mathrm{i}\tau)}$, so $\Phi(\mathrm{i}\tau) \in \mathbb{R}$ for $|\tau| < \pi/8$, with $\Phi(0) \approx 0.446697 > 0$. Evaluating the absolutely convergent series (1.2) directly at $\tau = 0.31$ and $\tau = 0.35$ (where $\cos(4\tau) > 0.169 > 0$ gives rapid geometric decay) yields $\Phi(0.31\mathrm{i}) \approx +1.892333 > 0$ and $\Phi(0.35\mathrm{i}) \approx -13.866622 < 0$ (reflecting the sharp drop of $f(\mathrm{i}\tau) = \mathrm{e}^{\mathrm{i}\tau}\theta(\mathrm{e}^{4\mathrm{i}\tau}) \in \mathbb{R}$ toward 0 as $\mathrm{e}^{4\mathrm{i}\tau} \to \mathrm{i}$, where $\theta_4(\varepsilon) = 2\varepsilon^{-1/2}\sum_{m=0}^{\infty}\mathrm{e}^{-\pi(m+1/2)^2/\varepsilon} \to 0$). By the intermediate value theorem, $\Phi(\mathrm{i}\tau_0) = 0$ for some $\tau_0 \in (0.31, 0.35)$, refined to $\tau_0 \approx 0.3194150268$. The off-axis zero $u_1 \approx 0.1278237567 + 0.3477699161\mathrm{i}$ ($|\Phi'(u_1)| \approx 1593.82$) is certified via Cauchy's argument principle on $|u - u_1| = 0.01$ inside $\mathcal{U}_{\pi/8}$. □

Remark 3.3 (Boundary $|\eta| = \pi/4$ and logarithmic singularities of $\ln G_y$). Two cautions are in order. First, while $w = \mathrm{i}y$ ($y \in \mathbb{R}$) is classically a natural boundary of $\theta(w)$ due to its dense modular cusps at rational y, the mere fact that $|\exp(-\pi n^2\mathrm{e}^{2L}\mathrm{i}\sin 2\eta)| = 1$ at $\cos(2\eta) = 0$ only shows divergence of the defining series (1.2), not non-continuability by itself (indeed along the single ray $u = \mathrm{i}\tau \to \mathrm{i}\pi/8$, $w = \mathrm{e}^{4\mathrm{i}\tau} \to \mathrm{i}$ hits a cusp where $\Phi(\mathrm{i}\tau) \to 0$). Second, although zeros of $G_y(z)$ are harmless for the Cauchy shift (1.6) of the holomorphic integrand $G_y(z)\mathrm{e}^{2\mathrm{i}tz}$, every zero $G_y(z_0) = 0$ in $\mathcal{S}_{\pi/4}$ is a *logarithmic branch point* of $\ln G_y(z)$ and a *simple*

pole of $\partial_z \ln G_y(z)$, obstructing naive global steepest-descent arguments that treat $\ln G_y(z)$ as holomorphic throughout $\mathcal{S}_{\pi/4}$.

4. Conditional local saddle-point coalescence normal form

Away from the zero locus $\{z \in \mathcal{S}_{\pi/4} : G_y(z) = 0\}$, $\ln G_y(z)$ is locally holomorphic and complex stationary points of $\Psi_{y,\omega}(z) := \ln G_y(z) + \mathrm{i}\omega z$ ($\omega = 2t > 0$) satisfy $\partial_z \ln G_y(z) + \mathrm{i}\omega = 0$. We record the exact local bifurcation normal form under explicit hypotheses.

Lemma 4.1 (Imaginary-axis derivative parity away from Φ-zeros)**.** *Let $y \in \mathbb{R}$ and $\eta \in (-\pi/4, \pi/4)$ satisfy $G_y(\mathrm{i}\eta) > 0$, and write $L_y(\eta) := \ln G_y(\mathrm{i}\eta) \in \mathbb{R}$. Then on a neighborhood of $z = \mathrm{i}\eta$:*

(a) *Every odd horizontal derivative $\partial_\xi^{2k+1} \ln G_y(\mathrm{i}\eta)$ is purely imaginary and every even horizontal derivative $\partial_\xi^{2k} \ln G_y(\mathrm{i}\eta)$ is purely real; in particular,*

$$\partial_z \ln G_y(\mathrm{i}\eta) = -\mathrm{i}L_y'(\eta), \qquad \partial_z^2 \ln G_y(\mathrm{i}\eta) = \partial_\xi^2 \ln G_y(\mathrm{i}\eta) = -L_y''(\eta), \qquad \partial_z^3 \ln G_y(\mathrm{i}\eta) = \mathrm{i}L_y'''(\eta). \tag{4.1}$$

(b) *If $z_* = \xi_* + \mathrm{i}\eta_*$ ($G_y(z_*) \neq 0$) solves $\partial_z \ln G_y(z_*) + \mathrm{i}\omega = 0$, then $-\bar{z}_* = -\xi_* + \mathrm{i}\eta_*$ also solves $\partial_z \ln G_y(-\bar{z}_*) + \mathrm{i}\omega = 0$.*

Proof. Since $G_y(\mathrm{i}\eta) > 0$, the real-valued branch $\ln G_y(\mathrm{i}\eta) = 2\ln|\Phi((y+\mathrm{i}\eta)/2)|$ extends holomorphically to a disk $D(\mathrm{i}\eta, r)$ on which $G_y(z) \neq 0$, preserving $\ln G_y(-\xi + \mathrm{i}\eta) = \overline{\ln G_y(\xi + \mathrm{i}\eta)}$. Differentiating m times with respect to ξ at $\xi = 0$ gives $(-1)^m \partial_\xi^m \ln G_y(\mathrm{i}\eta) = \overline{\partial_\xi^m \ln G_y(\mathrm{i}\eta)}$, proving the reality/imaginarity claim, while holomorphy in $z = \xi + \mathrm{i}\eta$ ($\partial_z = -\mathrm{i}\partial_\eta$) yields (4.1). Differentiating $\ln G_y(-\bar{z}) = \overline{\ln G_y(z)}$ gives $-\overline{\partial_z \ln G_y(-\bar{z})} = \overline{\partial_z \ln G_y(z)}$, so $\partial_z \ln G_y(z_*) = -\mathrm{i}\omega$ implies $\partial_z \ln G_y(-\bar{z}_*) = \overline{\mathrm{i}\omega} \cdot (-1) \cdot (-1) = -\mathrm{i}\omega$. □

Proposition 4.2 (Local 90° Stokes-rotation saddle coalescence normal form)**.** *Fix $\omega > 0$ and assume $(y_c, \eta_c) \in (0,\infty) \times (0, \pi/4)$ satisfies $G_{y_c}(\mathrm{i}\eta_c) > 0$ together with*

$$L_{y_c}'(\eta_c) = \omega, \qquad L_{y_c}''(\eta_c) = 0, \qquad \alpha := \partial_y L_{y_c}'(\eta_c) > 0, \qquad \beta := -L_{y_c}'''(\eta_c) > 0. \tag{4.2}$$

Then there exist $\delta_y > 0$ and $\delta_z > 0$ such that for $|y - y_c| < \delta_y$, the solutions of $\partial_z \ln G_y(z) + \mathrm{i}\omega = 0$ in $|z - \mathrm{i}\eta_c| < \delta_z$ consist of two roots colliding quadratically at $\mathrm{i}\eta_c$:

(i) *for $y_c - \delta_y < y < y_c$, a symmetric horizontal off-axis pair $z_*^\pm(y,\omega) = \pm\xi_*(y,\omega) + \mathrm{i}\eta_*(y,\omega)$ satisfying*

$$\boxed{\left(z_*^\pm(y,\omega) - \mathrm{i}\eta_c\right)^2 = K^2(y_c - y) + \mathcal{O}\left(|y - y_c|^{3/2}\right), \qquad K := \sqrt{\frac{2\,\partial_y L_{y_c}'(\eta_c)}{-L_{y_c}'''(\eta_c)}} > 0,} \tag{4.3}$$

so $\xi_*(y,\omega) = K\sqrt{y_c - y} + \mathcal{O}(|y - y_c|) > 0$ and $\eta_*(y,\omega) = \eta_c + \mathcal{O}(|y - y_c|)$;

(ii) *for $y_c < y < y_c + \delta_y$, two purely imaginary roots $z_*^\pm(y,\omega) = \mathrm{i}\eta_\pm(y,\omega)$ with $\eta_\pm(y,\omega) = \eta_c \pm K\sqrt{y - y_c} + \mathcal{O}(|y - y_c|)$, of which the lower branch $\eta_*(y,\omega) := \eta_-(y,\omega) < \eta_c$ carries strictly negative horizontal Hessian $\partial_\xi^2 \ln G_y(\mathrm{i}\eta_*) = -\sqrt{2\alpha\beta(y - y_c)} + \mathcal{O}(|y - y_c|) < 0$.*

Proof. Since $G_{y_c}(\mathrm{i}\eta_c) > 0$, $(y,z) \mapsto F(y,z) := \partial_z \ln G_y(z) + \mathrm{i}\omega$ is holomorphic on a bidisk $\{|y - y_c| < r_1\} \times \{|z - \mathrm{i}\eta_c| < r_2\}$. Setting $\Delta z := z - \mathrm{i}\eta_c = \xi + \mathrm{i}\Delta\eta$ and $\Delta y := y - y_c$, Lemma 4.1(a) and (4.2) yield the Taylor expansion

$$F(y,z) = -\mathrm{i}\alpha\,\Delta y - \frac{\mathrm{i}\beta}{2}(\Delta z)^2 + c_{11}\,\Delta y\,\Delta z + c_{03}(\Delta z)^3 + \mathcal{O}\left(|\Delta y|^2 + |\Delta y||\Delta z|^2 + |\Delta z|^4\right).$$

By the Weierstrass preparation theorem for the double root $\Delta z = 0$ at $\Delta y = 0$, $F(y,z) = U(y,z)\left[(\Delta z)^2 + a_1(\Delta y)\Delta z + a_0(\Delta y)\right]$ on a smaller bidisk, where $U(y_c, \mathrm{i}\eta_c) = -\mathrm{i}\beta/2 \neq 0$. The reflection symmetry $F(y, -\bar{z}) = -\overline{F(y,z)}$ forces the unique monic Weierstrass polynomial $P(\Delta y, \Delta z)$ to satisfy $\overline{P(\Delta y, -\overline{\Delta z})} = P(\Delta y, \Delta z)$, whence $a_1(\Delta y) \in \mathrm{i}\mathbb{R}$ with $a_1(\Delta y) = \mathcal{O}(|\Delta y|)$ and $a_0(\Delta y) \in \mathbb{R}$ with $a_0(\Delta y) = \frac{2\alpha}{\beta}\Delta y + \mathcal{O}(|\Delta y|^2) = -K^2(y_c - y) + \mathcal{O}(|\Delta y|^2)$. Solving $(\Delta z + \frac{1}{2}a_1(\Delta y))^2 = \frac{1}{4}a_1(\Delta y)^2 - a_0(\Delta y) = K^2(y_c - y) + \mathcal{O}(|\Delta y|^2)$ gives the conjugate-reflected off-axis pair (4.3) for $y < y_c$ (since $-\frac{1}{2}a_1(\Delta y) \in \mathrm{i}\mathbb{R}$) and two purely imaginary roots

$\Delta z = -\frac{1}{2}a_1(\Delta y) \pm \mathrm{i}K\sqrt{y-y_c} + \mathcal{O}(|y-y_c|) \in \mathrm{i}\mathbb{R}$ for $y > y_c$. Evaluating $\partial_z F(y,z) = -\mathrm{i}\beta\,\Delta z + \mathcal{O}(|\Delta y| + |\Delta z|^2)$ at the lower imaginary root $\Delta z = -\mathrm{i}K\sqrt{y-y_c} + \mathcal{O}(|y-y_c|)$ gives $\partial_\xi^2 \ln G_y(\mathrm{i}\eta_*) = -\beta K\sqrt{y-y_c} + \mathcal{O}(|y-y_c|) = -\sqrt{2\alpha\beta(y-y_c)} + \mathcal{O}(|y-y_c|) < 0$. □

Remark 4.3 (Local theorem versus global numerical observation). Proposition 4.2 is strictly *local* near $(y_c, \mathrm{i}\eta_c)$: as shown in the proof, the quadratic normal form $(\Delta z)^2 \approx K^2(y_c - y)$ is actually a complex saddle-node / 90° Stokes-rotation coalescence in which two horizontal off-axis saddles ($\pm\xi_* + \mathrm{i}\eta_*$ for $y < y_c$) collide at $\mathrm{i}\eta_c$ and separate vertically along the imaginary axis into two purely imaginary saddles ($\mathrm{i}\eta_\pm$ for $y > y_c$, of which the lower branch $\eta_* = \eta_- < \eta_c$ has negative horizontal Hessian $\partial_\xi^2 \ln G_y(\mathrm{i}\eta_*) < 0$). Global behavior across $y \in [0.05, 1.10]$ at $\omega = 2\gamma_1$ is documented as a numerical homotopy observation in Section 5, not claimed as an unconditional global theorem for all $\omega > 0$.

5. High-precision verification and quadrature conditioning

All computations are reproduced by the standalone verification script `scripts/verify_paper1_wigner_saddle_coalescence.py` using `mpmath` at 50–60 decimal places.

5.1. **Cauchy wall decay and closed-loop verification.** At $y = 0.40$, $z = 0.35 + 0.55\mathrm{i}$, and $u = 0.28 + 0.19\mathrm{i}$, `mpmath` (60 dps) confirms $|\Phi(u) - \Phi(-u)| = 2.61 \times 10^{-61}$ and $|G_y(-\xi + \mathrm{i}\eta) - \overline{G_y(\xi + \mathrm{i}\eta)}| = 1.17 \times 10^{-59}$. Table 2 shows the vertical wall decay (3.1) up to the strip boundary $\eta = \pi/4$. On $[-2.4, 2.4] \times [0, 0.55]$ at $\omega = \gamma_1$, the closed Cauchy contour residual is 3.59×10^{-43}.

Table 2. Vertical Cauchy wall magnitude $|G_y(L + \mathrm{i}\eta)|$ ($y = 0.40$, 60-decimal `mpmath`) versus contour height $\eta \in [0, \pi/4]$.

Height η	$\cos(2\eta)$	$\lvert G_y(2.0 + \mathrm{i}\eta)\rvert$	$\lvert G_y(3.0 + \mathrm{i}\eta)\rvert$	**Series / Wall Regime**
0.000000	1.000000	1.38092×10^{-189}	$9.77371 \times 10^{-1459}$	Convergent (decays)
0.250000	0.877583	3.42020×10^{-165}	$1.69241 \times 10^{-1278}$	Convergent (decays)
0.500000	0.540302	5.53601×10^{-98}	6.53404×10^{-782}	Convergent (decays)
0.550000	0.453596	1.04943×10^{-80}	2.98538×10^{-654}	Convergent (decays)
0.650000	0.267499	1.27103×10^{-43}	2.95791×10^{-380}	Convergent (decays)
0.750000	0.070737	2.05308×10^{-4}	1.47351×10^{-90}	Convergent (decays)
$\pi/4 \approx 0.785398$	0.000000	Series diverges	Series diverges	Strip boundary ($\cos 2\eta = 0$)

5.2. **Numerical saddle-branch tracking at $\omega = 2\gamma_1$.** At $\omega_1 = 2\gamma_1 \approx 28.2694502835$, solving $L_y'(\eta) = \omega_1$ and $L_y''(\eta) = 0$ simultaneously yields

$$y_c(\omega_1) = 0.43202053030568070438\ldots, \qquad \eta_c(\omega_1) = 0.60271502385742573679\ldots,$$

where $G_{y_c}(\mathrm{i}\eta_c) \approx 82.3717 > 0$, $\partial_y L'_{y_c}(\eta_c) \approx +48.0778 > 0$, $L'''_{y_c}(\eta_c) \approx -1074.1651 < 0$, and $K \approx 0.29919336$, verifying all hypotheses of Proposition 4.2 at ω_1. Table 3 lists the numerically tracked branch across $y \in [0.05, 1.10]$.

5.3. **Quadrature conditioning and canonical normalization verification.** At $y = 0.40$ and $\omega = 2\gamma_1$, the net integral $I(0.40, \gamma_1) = +1.76903968 \times 10^{-6}$ is invariant across $\eta \in [0, 0.60]$. Shifting from $\eta = 0$ (where $I_+ = +7.6829 \times 10^{-4}$, $I_- = -7.6653 \times 10^{-4}$, and $\kappa_{\mathrm{quad}} = 867.60$) to $\eta = 0.50$ (where $I_+ = +1.7697 \times 10^{-6}$, $I_- = -6.9457 \times 10^{-10}$, and $\kappa_{\mathrm{quad}} = 1.0008$) strongly improves numerical quadrature conditioning. Table 4 verifies both $\int_0^\infty I(y, \gamma_k)\,\mathrm{d}y = 0$ and the canonical prefactor 16 in (1.5) against direct Cauchy–Riemann evaluation $W_{\mathrm{CR}}(0.20, \gamma_k)$ to 54–58 decimals across $\gamma_1, \ldots, \gamma_5$.

TABLE 3. 50-decimal tracked stationary-point branch $z_*^{\pm}(y, 2\gamma_1) = \pm\xi_* + \mathrm{i}\eta_*$ across $y \in [0.05, 1.10]$ (residuals $< 10^{-45}$).

Shift y	**Right Saddle** $+\xi_* + \mathrm{i}\eta_*$	**Hessian** $\mathrm{Re}\,\partial_z^2 \ln G_y(z_*^+)$	**Tracked Regime**
0.050000	$+0.40418674 + 0.61381428\mathrm{i}$	$-4.54229932 \times 10^{1}$	Off-axis ($\xi_* > y$)
0.100000	$+0.41411144 + 0.60599972\mathrm{i}$	$-3.22653864 \times 10^{1}$	Off-axis ($\xi_* > y$)
0.200000	$+0.33379542 + 0.58444323\mathrm{i}$	$-8.54838506 \times 10^{1}$	Off-axis ($\xi_* > y$)
0.300000	$+0.13969227 + 0.62533792\mathrm{i}$	$-7.55807305 \times 10^{1}$	Off-axis ($0 < \xi_* < y$)
0.350000	$+0.15981952 + 0.60988092\mathrm{i}$	$-5.93841563 \times 10^{1}$	Off-axis ($0 < \xi_* < y$)
0.400000	$+0.03097628 + 0.63405509\mathrm{i}$	$-2.28083509 \times 10^{1}$	Near-axis off-axis
0.420000	$+0.02826368 + 0.61080112\mathrm{i}$	$-1.39099980 \times 10^{1}$	Near-axis off-axis
0.432021	$+0.00000000 + 0.60271502\mathrm{i}$	$0\ (-1.03 \times 10^{-48})$	Local coalescence $y_c(2\gamma_1)$
0.450000	$+0.00000000 + 0.54821071\mathrm{i}$	$-2.67260091 \times 10^{1}$	Lower on-axis branch ($\xi_* = 0$)
0.500000	$+0.00000000 + 0.46989397\mathrm{i}$	$-4.08129949 \times 10^{1}$	Lower on-axis branch ($\xi_* = 0$)
0.600000	$+0.00000000 + 0.36453529\mathrm{i}$	$-6.30701027 \times 10^{1}$	Lower on-axis branch ($\xi_* = 0$)
0.750000	$+0.00000000 + 0.26030095\mathrm{i}$	$-9.85437766 \times 10^{1}$	Lower on-axis branch ($\xi_* = 0$)
0.900000	$+0.00000000 + 0.18954242\mathrm{i}$	$-1.41913558 \times 10^{2}$	Lower on-axis branch ($\xi_* = 0$)
1.100000	$+0.00000000 + 0.12569397\mathrm{i}$	$-2.20145746 \times 10^{2}$	Lower on-axis branch ($\xi_* = 0$)

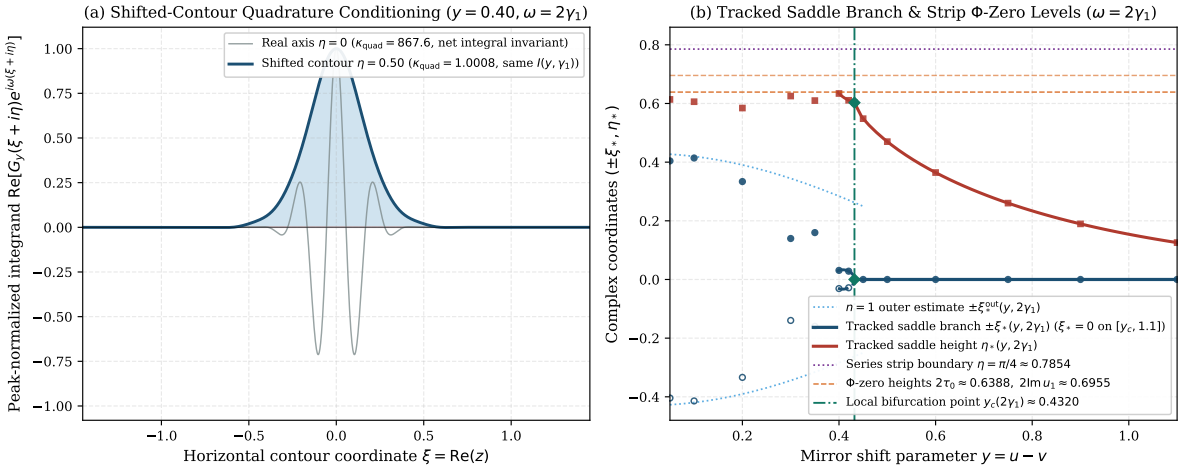


FIGURE 1. **(a)** Peak-normalized integrand $\mathrm{Re}[G_y(\xi + \mathrm{i}\eta)\mathrm{e}^{\mathrm{i}\omega(\xi+\mathrm{i}\eta)}]$ at $y = 0.40$, $\omega = 2\gamma_1$: the net integral $I(y, \gamma_1) = 1.76903968 \times 10^{-6}$ is contour-invariant, while shifting from $\eta = 0$ to $\eta = 0.50$ lowers the L^1 quadrature condition number κ_{quad} from 867.6 to 1.0008. **(b)** Numerically tracked stationary-point branch $(\pm\xi_*, \eta_*)$ at $\omega = 2\gamma_1$ alongside the strip Φ-zero heights $2\tau_0 \approx 0.6388$ and $2\,\mathrm{Im}(u_1) \approx 0.6955$.

TABLE 4. 50-decimal verification of the zero-marginal identity $\int_0^\infty I(y, \gamma_k)\,\mathrm{d}y = 0$, canonical Wronskian normalization $W_{\mathrm{Wig}}(0.20, \gamma_k) = 16\int_0^\infty y \sinh(0.40y) I(y, \gamma_k)\,\mathrm{d}y$ versus exact $W_{\mathrm{CR}}(0.20, \gamma_k)$, and sign-change count of $I(y, \gamma_k)$ on $[0, 1.35]$.

Zero	γ_k	$\int_0^\infty I(y, \gamma_k)\,\mathrm{d}y$	$W_{\mathrm{Wig}}(0.20, \gamma_k)$	$W_{\mathrm{CR}}(0.20, \gamma_k)$	$\lvert W_{\mathrm{Wig}} - W_{\mathrm{CR}}\rvert$	**Crossings**
γ_1	14.134725	-3.68×10^{-54}	$+3.8445657091 \times 10^{-7}$	$+3.8445657091 \times 10^{-7}$	7.07×10^{-55}	1
γ_2	21.022040	$+9.22 \times 10^{-55}$	$+6.3713228683 \times 10^{-11}$	$+6.3713228683 \times 10^{-11}$	9.85×10^{-58}	1
γ_3	25.010858	$+1.07 \times 10^{-54}$	$+3.2438994312 \times 10^{-13}$	$+3.2438994312 \times 10^{-13}$	2.09×10^{-55}	5
γ_4	30.424876	-1.54×10^{-53}	$+1.1892708589 \times 10^{-16}$	$+1.1892708589 \times 10^{-16}$	2.15×10^{-54}	6
γ_5	32.935062	-1.32×10^{-54}	$+3.4236969843 \times 10^{-18}$	$+3.4236969843 \times 10^{-18}$	2.45×10^{-55}	7

6. SCOPE, LIMITATIONS, AND EXPLICIT OBSTRUCTIONS

To delineate strictly what the phase-space and contour formalism can and cannot establish, we record four explicit obstructions:

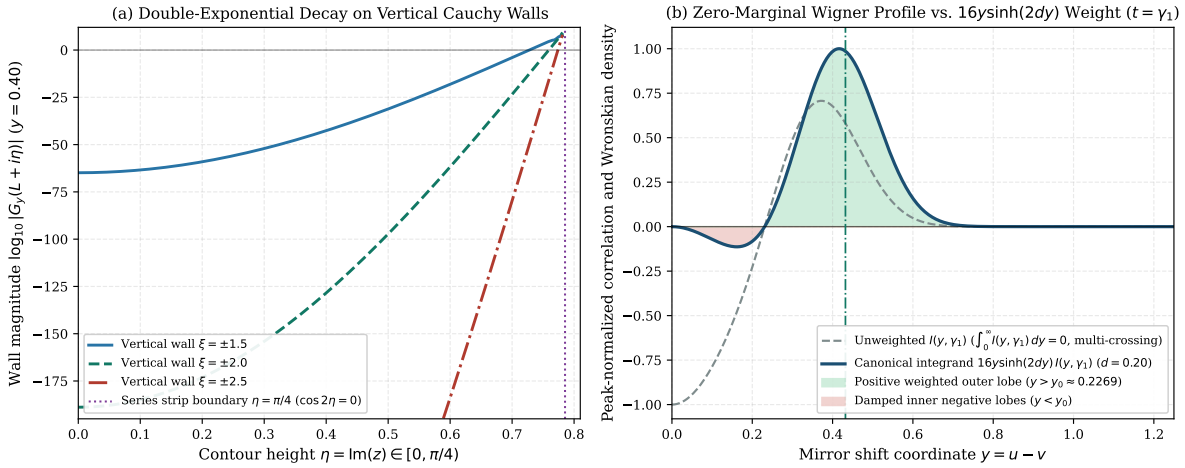


FIGURE 2. **(a)** Double-exponential decay $\log_{10}|G_y(L+\mathrm{i}\eta)|$ on vertical walls $\xi=\pm L$ $(y=0.40)$ up to the series strip boundary $\eta=\pi/4$. **(b)** Unweighted zero-marginal profile $I(y,\gamma_1)$ versus the canonical weighted integrand $16y\sinh(2dy)I(y,\gamma_1)$ at $d=0.20$.

(1) **Contour invariance versus sign proof:** By Cauchy's theorem, (1.6) equals $I(y,t)$ identically for every $\eta\in(0,\pi/4)$. Shifting η reduces the L^1 norm of the integrand (conditioning numerical quadrature), but cannot turn a negative integral positive; indeed, $I(y,\gamma_1)<0$ for $y\in(0,0.2269)$, where shifted contours simply exhibit negative dominant lobes.

(2) **Zeros of $\Phi(u)$ and steepest-descent topology inside $\mathcal{S}_{\pi/4}$:** Because $\Phi(\mathrm{i}\tau_0)=0$ at $\tau_0\approx0.319415<\pi/8$ and $\Phi(u_1)=0$ at $u_1\approx0.127824+0.347770\mathrm{i}$ ($2\tau_0\approx0.638830$ and $2\,\mathrm{Im}\,u_1\approx0.695540$, both inside $\mathcal{S}_{\pi/4}$), $G_y(z)$ has zeros inside $\mathcal{S}_{\pi/4}$ creating logarithmic branch points in $\ln G_y(z)$ above $\eta_c(2\gamma_1)\approx0.602715$ (which the tracked saddle height approaches for small y and large ω). While horizontal contours $z=\xi+\mathrm{i}\eta$ remain valid throughout $|\eta|<\pi/4$ by holomorphy of $G_y(z)$, any global steepest-descent deformation for $\exp(\ln G_y(z)+\mathrm{i}\omega z)$ across $\mathcal{S}_{\pi/4}$ must account for Stokes lines around logarithmic sinks $G_y(z_0)=0$.

(3) **Multi-crossing oscillations as t grows:** For $k\geq3$, $I(y,\gamma_k)$ changes sign multiple times (Table 4), so Karlin's single-crossing sign variation lemma does not yield $W(d,\gamma_k)>0$.

(4) **Davenport–Heilbronn and backward heat-flow obstruction:** Every identity in Proposition 2.1 uses only real-evenness and rapid decay of $\Phi(u)$, never the Euler product of $\zeta(s)$. If Φ is replaced by $\Phi_\lambda(u)=\mathrm{e}^{\lambda u^2}\Phi(u)$ with $\lambda<0$ (backward de Bruijn–Newman flow H_λ [4, 9, 2]) or by the kernel of the Davenport–Heilbronn function [3, 14], then $W_\lambda(d,t)=16\int_0^\infty y\sinh(2dy)I_\lambda(y,t)\,\mathrm{d}y$ has the *exact same* Wigner–sinh$(2dy)$ form, yet H_λ $(\lambda<0)$ and Davenport–Heilbronn possess infinitely many off-critical zeros where $W<0$ nearby. Hence phase-space and saddle analysis of Φ alone cannot exclude off-critical zeros without arithmetic input distinguishing $\zeta(s)$ from non-Euler-product Dirichlet series.

APPENDIX A. HEURISTIC SINGLE-MODE $n=1$ SADDLE SCALES

Retaining only the $n=1$ term $\phi_1(u)\approx2\pi^2\mathrm{e}^{9u}\exp(-\pi\mathrm{e}^{4u})$ for u with large positive real part gives two heuristic comparison scales used in Figure 1(b):

(1) For $\xi>y\geq0$, $\partial_z\ln G_y^{(1)}(z)\approx9-4\pi\cosh(2y)\mathrm{e}^{2z}$, so $\partial_z\ln G_y^{(1)}(z)+\mathrm{i}\omega=0$ yields $\mathrm{e}^{2z_*}=\frac{9+\mathrm{i}\omega}{4\pi\cosh(2y)}$, giving $\eta_*^{\mathrm{out}}(\omega)=\frac{1}{2}\arctan(\omega/9)$ $(\approx0.631289<\pi/4$ at $\omega=2\gamma_1)$ and $\xi_*^{\mathrm{out}}(y,\omega)=\frac{1}{2}\ln\frac{\sqrt{\omega^2+81}}{4\pi\cosh(2y)}$.

(2) In the heuristic regime $0\leq|\xi|\ll y$ where $\Phi((z-y)/2)$ varies more slowly than $\Phi((z+y)/2)$, $\partial_z\ln G_y^{(1)}(z)\approx\frac{9}{2}-2\pi\mathrm{e}^{y+z}$, giving the rough coalescence scale $y_c^{\mathrm{inner}}(\omega)\approx$

$\frac{1}{2}\ln(\frac{\omega}{4\pi}) \approx 0.405379$ at $\omega = 2\gamma_1$ (within 6.2% of the exact root $y_c(2\gamma_1) \approx 0.432021$). Because $n \geq 2$ theta modes and the subleading $-3\pi e^{5u}$ term are non-negligible near $u \approx 0$, these $n = 1$ expressions serve purely as heuristic guides rather than asymptotic theorems at finite height.

Use of AI and Computational Tools

Portions of the symbolic algebra exploration, literature cross-checking, LaTeX manuscript preparation, and high-precision Python/`mpmath` verification script (`scripts/verify_paper1_wigner_saddle_coalescence.py`) were assisted by **Gemini** (Google DeepMind). All mathematical proofs, normalizations, counterexample checks, and numerical outputs were independently verified at 50–60 decimal places.

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